tings can be saved as profiles and can be loaded from the hard drives—a feature now available with all routers.

Performance

Zone 1

In zone one, a wooden partition was the only barrier between the laptop and WiFi router. As shown by PassMark WirelessMon, the signal strength on all routers hovered in the range of 65-70 per cent, with only the D-link DIR-615 falling slightly below 60 per cent.

In the 'g' group, almost all routers took nearly 3 minutes to transfer a single 400 MB file, and that's barely acceptable performance. On the brighter side, streaming a single movie posed no problems. However, multiple streams led to framing. In the multiple file copy tests too, the time taken for copy didn't deviate too much. Again, most routers completed it in 3 minutes. The Compex NP25G lagged behind other 802.11g routers by nearly 20 seconds.

In the 'n' group, the routers were able to complete the file transfer in almost a minute. In both, the single file and multiple file transfer, the time hovered around 70 seconds, which is certainly not bad. Linksys' WRT310N was behind other routers, but not by large margins. We had absolutely no problems streaming a high definition 720p movie over n routers.

Zone 2

Zone two performances are more critical than zone one, because it truly reflects the real world scenario. Between the router and the client, we had two concrete walls and wooden cabinets. Also, they were in two different rooms, but on the same floor.

In the 'g' group, the signal reception immediately fell below 40 per cent and kept varying between 35-40 per cent. Despite the signal attenuation, there wasn't a significant difference in the performance during file transfers. In fact, most routers took just 5-7 seconds more to complete the test. However, the same can't be said about movie streaming. Almost all routers played the movie, but we could notice frames dropping. Compex's NP54G was the only router where the viewing experience was hampered.

In the 'n' group, the signal reception dropped too. However, the likes of D-link DIR-655 managed to stay at around 60 per cent—good. In the file transfer test, the 802.11n-based routers were able to deliver the throughput despite lower signal strength. Movie streaming was smooth, and all of them coped well with multiple streams as well.

From the results, it is quite clear that the n-based routers do have an edge. However, there are some compatibility issues to consider. When we used a laptop with an 802.11g wireless card, the performance wasn't quite up to the mark, and the figures were too low.

Jargon Buster

SSID—SSID or Service Set Identifier (Network Name) is the name (alphanumeric) given to a Wireless Local Area network (WLAN). To communicate with any device on a wireless network, all wireless devices must use the same SSID. The SSID can be kept private, in which case, a user who doesn't know the SSID cannot gain access to that network.

WEP—Wired Equivalent Privacy. As the name suggests, this was the earliest 802.11 standard, providing 64- and 128-bit keys (hexadecimal coded), to protect wireless devices from unwanted intrusion.

WPA/WPA2—WPA is an acronym for Wireless Protected Access, which is a slightly more secure method of protecting data transmission than WEP. WPA 2 is a newer protocol. It's also known as 802.11i.

DHCP—Acronym for Dynamic Host Configuration Protocol. It's a communications protocol that automatically assigns IP addresses to clients logging on to a TCP/IP network.

Deep Packet Inspection—Deep Packet Inspection (DPI) is a form of packet filtering that inspects the actual data contained in data packets as opposed to just checking the header information. DPI is basically an evolved form of SPI (Stateful Packet Inspection).

The Winners

D-link's DIR-655 returned the best performance figures. Although based on 802.11n, it performed well with both 802.11n as well as 802.11g-based wireless cards. Also, it was the only router to have all the connectivity options that one can ask for. For its excellent performance, D-link's DIR-655 bags our best performance award.

If you are looking out for an 802.11n-based router that has a good combination of looks, features and performance, then the Linksys WRT310N is the one to choose. This router has all the traditional features offered by Linksys and is one of the routers with an easy-to-use interface. As a package, it offers a lot. A few might miss the USB option though. The Linksys WRT310N bags our *Best Buy* award.

Both Netgear products are attractively priced and hence make good buy if you are looking out for an easy to use WiFi router.

During the test we did encounter some compatibility issues, especially, when using the laptops legacy hardware with 802.11n-based devices. However, by setting the configuration correctly, we were able to derive better performance and range. The 802.11n standard definitely has a lot of potential and the current crop of draft 2.0 devices does offer some performance increment over the older standards. 802.11n should definitely be considered if you are planning a home WiFi network, especially with multimedia applications in mind. ■

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